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Data Science

I-Care: A Convolutional model for automated eye screening

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Abstract

Our project introduces an **AI-powered eye health assessment system** that offers affordable and accessible real-time screening. It uses CNNs on retinal images to detect diseases like retinopathy, cataracts, and macular degeneration; computer vision to track blink rate and eye redness for digital strain; and AI with Ishihara plates to identify types of color blindness. Designed to work on smartphones or webcams, it generates automated reports with personalized recommendations, enabling early detection, preventive care, and improved accessibility across communities.

Sustainable Development Goals(SDG) mapped

SDG 3: Good Health and Well-being Ensure healthy lives and promote well-being for all at all ages. Your project directly addresses early detection of eye diseases, preventive care, and accessibility to vision screening.

Introduction

- Eye health problems such as diabetic retinopathy, cataracts, macular degeneration, digital eye strain, and color vision deficiencies are becoming increasingly common in today's world.

- Factors such as long screen time, unhealthy lifestyles, and limited awareness contribute to rising cases of vision impairment.

- In real-life observations, we often see people—especially students, IT professionals, and elderly individuals—struggling with headaches, blurred vision, eye strain, or undiagnosed vision defects.

- Unfortunately, in rural and underserved communities, the lack of specialized ophthalmologists and affordable diagnostic tools makes early detection very difficult.

Motivation

- Many students and professionals ignore digital eye strain, while rural communities lack access to timely checkups for diseases like cataracts and retinopathy.
- Even color blindness often goes undiagnosed until later in life.
- These gaps in early detection inspired us to use AI and computer vision to build an affordable, scalable, and user-friendly eye health screening platform.

Objectives

The proposed system aims to address these issues by integrating multiple functionalities into one AI-powered platform:

1. **Disease Detection:** Use AI models (CNN) on retinal images to identify conditions such as diabetic retinopathy, cataracts, and macular degeneration.
2. **Color Vision Classification:** Employ Ishihara test images with AI models to detect and classify different types of color blindness.
3. **Digital Fatigue Analysis:** Apply computer vision (OpenCV) to fatigue rate, eye redness, and dryness using an image.
4. **Intelligent Assistance:** Uses a SerpAPI-powered chatbot to suggest nearby eye clinics or hospitals and guide users on next steps based on their results.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Early Detection of Child Eye Dryness Using IoT and Deep Learning Enabled AI Model	Maheswaran N., Bose S., Vijayaraj G., et al.	2025	LSTM IoT sensors for blink rate/env. data, real-time dryness detection	Focused on children, need broader and longer-term validation
2	Early Eye Disease Detection Using Machine Learning	Ashish Mishra et al.	2025	CNN on Kaggle eye images, GUI for live diagnosis	Small dataset 500 images), class imbalance
3	Eye Disease Detection using CNN	Dr. Praveenkumar A. Patel et al.	2025	CNNs on retinal images with preprocessing and augmentation	Dataset quality, interpretability, not replacement for doctor's diagnosis
4	Eye Disease Classification Using Transfer Learning and Explainable AI	Various authors, see included surveys	2023-25	Transfer Learning (ResNet, VGG, EfficientNet) and XAI for explanation	General gaps: model interpretability, clinical adoption, dataset diversity

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
5	Detection of Eye Disease in Retinal Images Based on Haar Wavelets	Urazmatov T., Djumanazarov O., Otamuratov K., Yusupova J.	2024	Urazmatov T., Djumanazarov O., Otamuratov K., Yusupova J.	Moderate accuracy ResNet: 86%, Tree: 78%
6	Diabetic Retinopathy Classification with Deep Learning via Fundus Images: A Short Survey	Zhu S., Xiong C., Zhong Q., Yao Y.	2024	Survey, ResNet, VGG, SqueezeNet, MobileNet on fundus datasets	Clinical deployment challenges, label inconsistencies
7	Exploration of Eye Fatigue Detection Features and Algorithm Based on Eye-Tracking Signal	Sun W., Wang Y., Hu B., Wang Q.	2024	Eye tracking, manual SVM, RF) and deep learning AE, CNN, Transformer)	Less real-world deployment, daily environment validation initially limited
8	Eye on the Future: Deep Learning with MobileNetV2 for Early Eye Disease Diagnosis	Shaheena Noor et al.	2024	MobileNetV2 on RGB fundus images, optimized for mobile/embedded devices	Resource-limited context focus; real-world deployment pending

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
9	Next-Gen Ophthalmic AI Early Eye Disease Detection with ResNet-18	Aakriti Kheterpal, Kanwarpartap Singh Gill	2024	ResNet-18 with fine tuning, image classification (cataract, DR, glaucoma, normal)	Lower performance for glaucoma cases; needs further refinement
10	Early Detection of Eye Disease in Retinal Images using Haar Wavelets and Decision Trees	Urazmatov T. et al.	2024	Haar Wavelet, decision trees on fundus images	Accuracy limited by method and dataset, not the highest in the field
11	An Efficient Deep Learning Model for Eye Disease Classification	Archana Saini, Kalpna Guleria, Shagun Sharma	2023	CNN (EfficientNetB3) with fundus images from Kaggle, batch size tuning	Generalizability, data diversity
12	Eye Health Monitoring System	Godhani K., Patel A., Shah H., Mehta A., Adhikari D.	2023	Blue light exposure monitoring using TCS34725 sensor and ML KNN, Naive Bayes)	Sensor-based physical parameters, not diagnostic; focuses on exposure not disease

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	Detection of Eye Strain Due to Usage of Electronic Devices: A Machine Learning Based Approach	Aggarwal D., Sharma D., Saxena A.B.	2022	Questionnaire + Random Forest classification	Based on subjective survey, not clinical diagnosis
14	Realtime recoloring Ishihara plates using Artificial Neural Networks for helping colorblind	Martín Montes, Alejandro Padilla, Juana Canul, Julio Ponce	2020	ANN-based recoloring for colorblind accessibility on Ishihara plates	Focus on accessibility, not detection, tested on limited subjects
15	Digital Eye Strain and Fatigue Recognition Using Electrooculogram Signals and Ultrasonic Distance Measurements	Robert K.C. Billones et al.	2018	EOG & ultrasonic sensors for detecting strain and optimal monitor distance	Small sample, limited demographic coverage

Research Gap(Limitations of existing systems)

- Most existing systems focus only on major eye diseases (diabetic retinopathy, glaucoma), ignoring others like dry eye syndrome and color vision deficiencies.
- Datasets used are often small and imbalanced, limiting model generalization and performance.
- Diagnosis relies mainly on retinal images, with little integration of multi-modal data such as patient history or real-time behavior.
- Lack of explainability reduces clinical trust in AI models.

Problem Definition

- Eye health disorders such as diabetic retinopathy, cataracts, macular degeneration, color blindness, and digital eye strain are prevalent worldwide and often go undiagnosed due to lack of accessible, affordable, and integrated screening solutions.
- Current methods either focus on a single issue (e.g., diabetic retinopathy detection or Ishihara color blindness test) or require specialized equipment and expertise, making them impractical for rural and underserved communities.

Scope

- Usable in **hospitals, schools, workplaces, and homes** for early eye disorder detection.
- Serves **students, professionals, elderly, and patients** with personalized tests & recommendations.
- **Customizable AI modules** integrate with healthcare platforms, telemedicine, and wearables.
- Provides **real-time analysis, preventive insights, and adaptive recommendations** to support long-term eye health.

Technological Stack

- **Disease Detection (CNN on Eye Images)**

Stack: TensorFlow/Keras (MobileNet/EfficientB3Net), Python, OpenCV
Method: Train CNN on eye disease classification with preprocessing (resize, normalize, augment).
Customization: Multi-disease detection (DR, Cataract, Glaucoma) with confidence-based results.

- **Color Vision Classification (Ishihara Test)**

Stack: Flask, OpenCV, Tesseract OCR, Scikit-learn
Method: Combines custom Ishihara-style tests with AI-based OCR and color analysis for automated response detection.
Simulation: Converts normal images to color-blind views (Protanopia, Deuteranopia, Tritanopia) using transformation matrices.

Technological Stack

- **Digital Eye Strain & Fatigue Analysis**

Stack: Python, MobileNetV2, Roboflow Pre trained model
Method: Detects redness, dryness, and fatigue using AI models; combines results with screen time and condition-based weights.
Computation: Calculates a unified fatigue score with adaptive recommendations.

- **Personalized Recommendation**

Stack: Python, Serp API, Groq API: Llama-3.3-70b-versatile
Method: Detects near by hosplitals with including chatbot
Computation: Combines user health metrics and location data with Serp API results and LLM-based reasoning to generate personalized healthcare recommendations.

Proposed system architecture/Working/Module Diagram

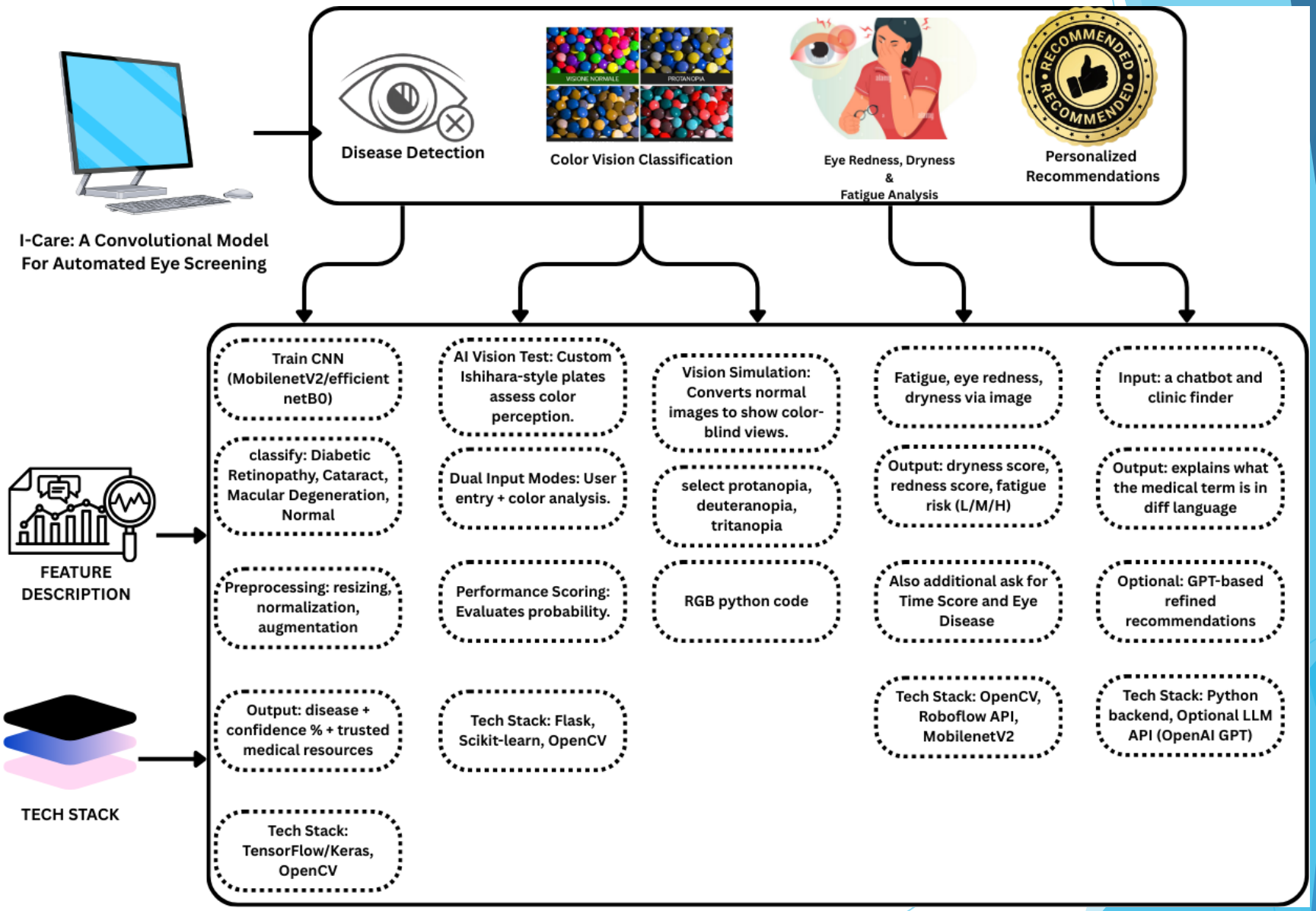

Figure 1: Proposed system architecture/Working/Module Diagram

Prototype Design Demonstration

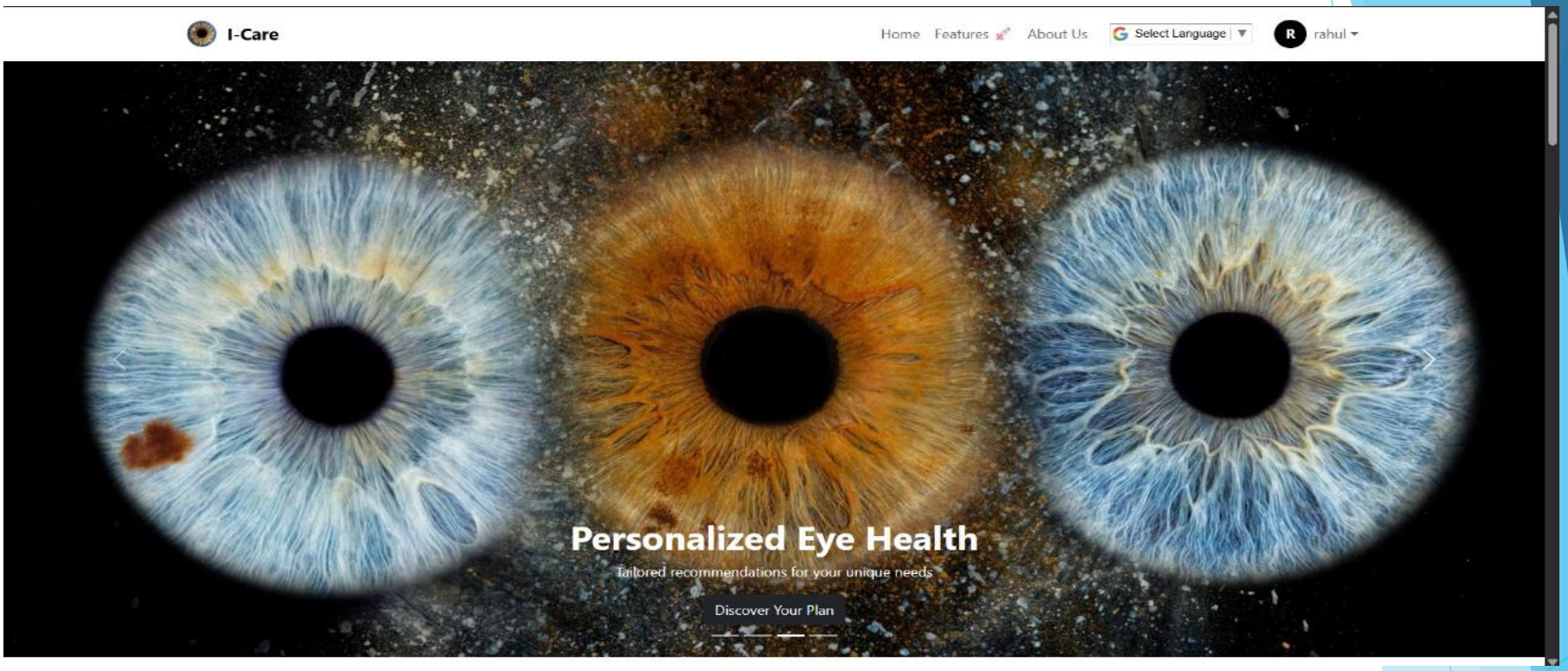

Figure 2: Homepage (From where we navigate to other features)

Implementation Status

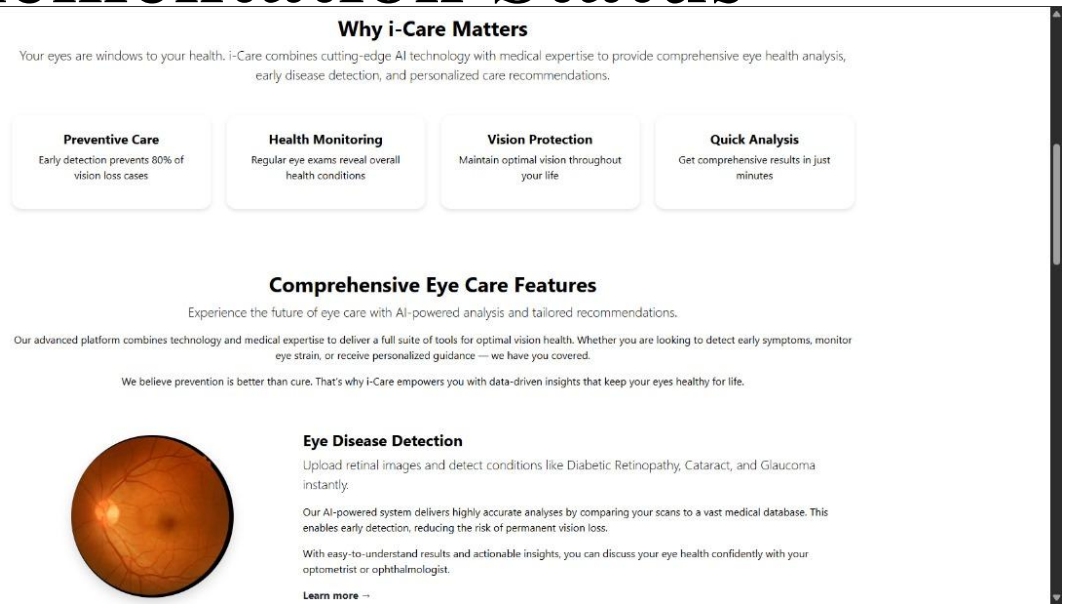

Figure 3: i-Care Platform – Home and Feature Overview

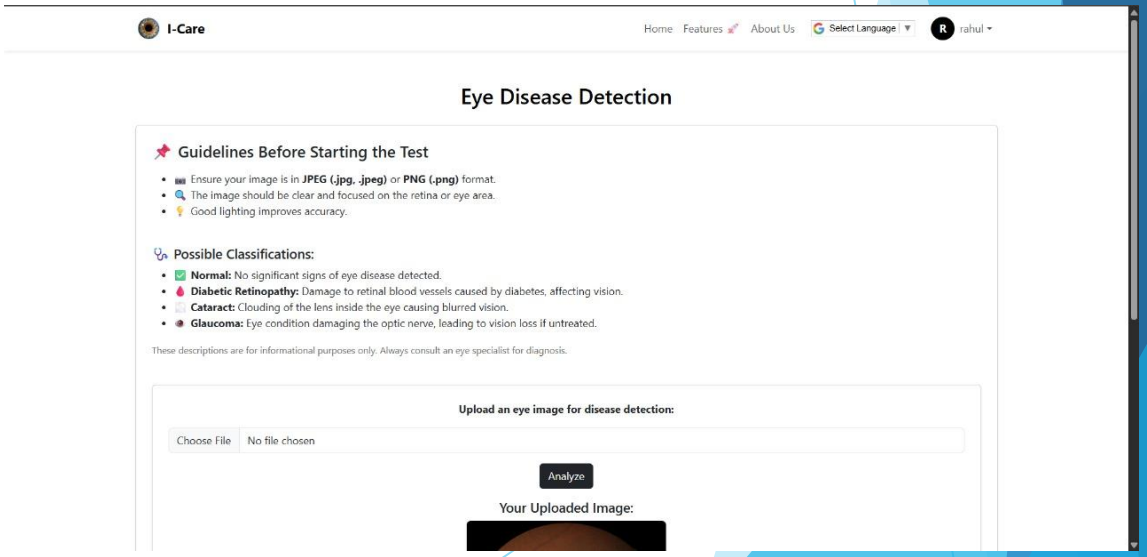

Figure 4: Eye Disease Detection – Upload Interface

Implementation Status

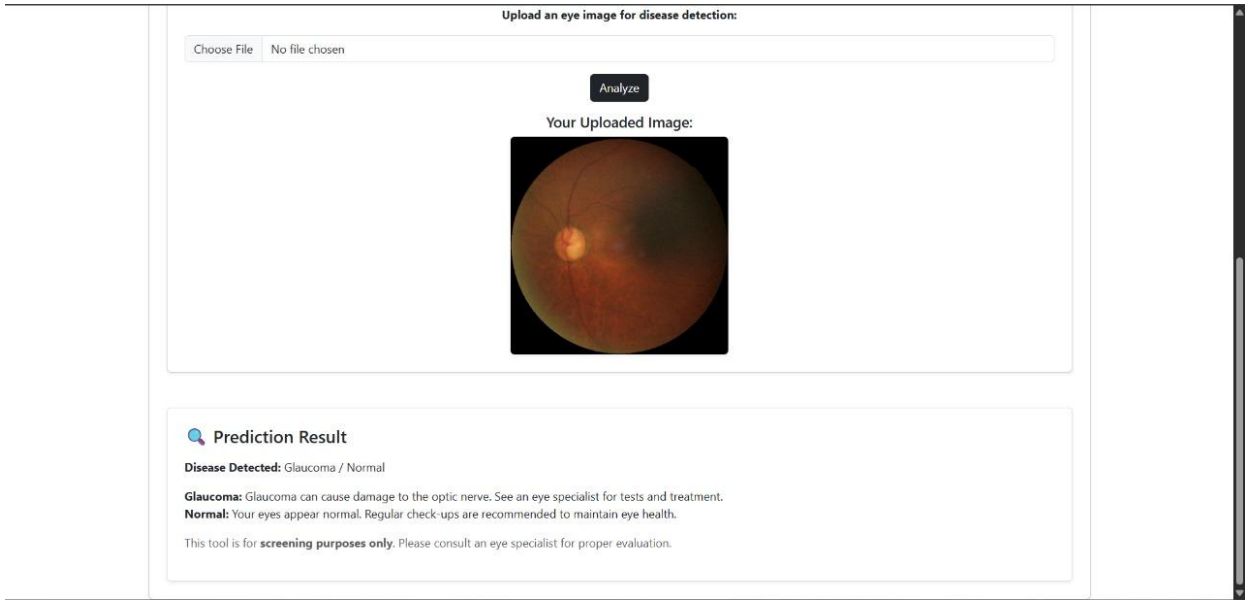

Figure 5: Prediction Results – Disease Detection Output

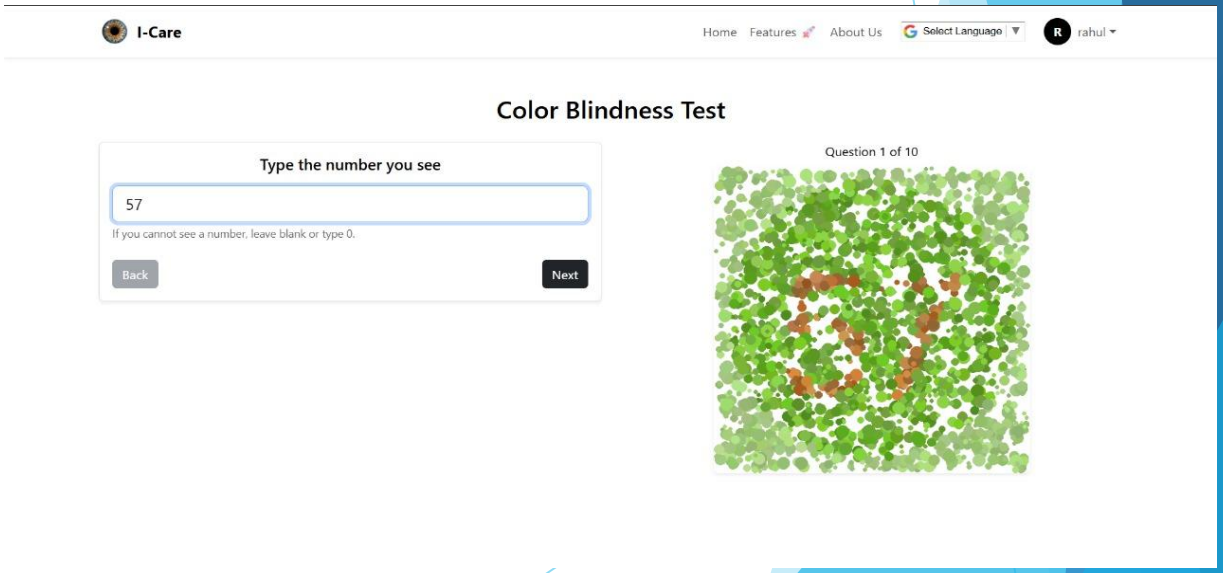

Figure 6: Color Blindness Test – User Interface

Implementation Status

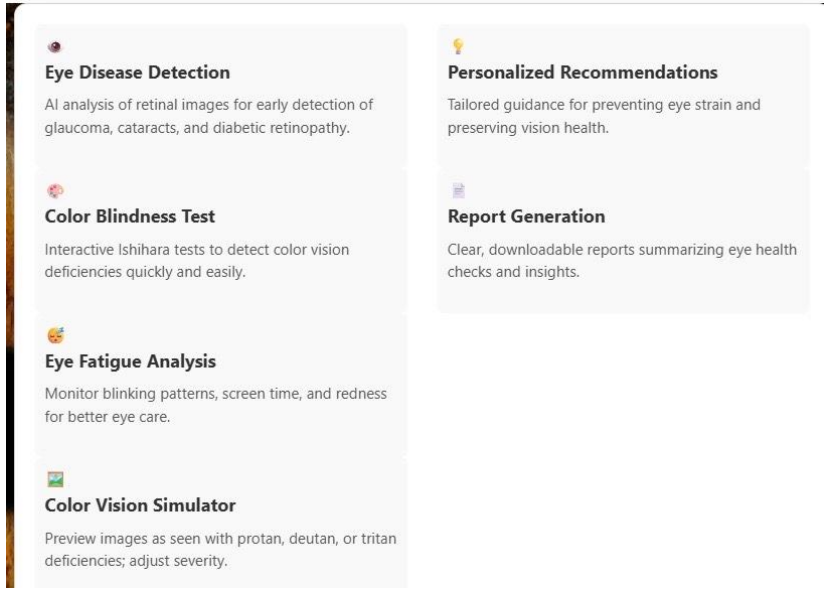

Figure 7: Features navigator

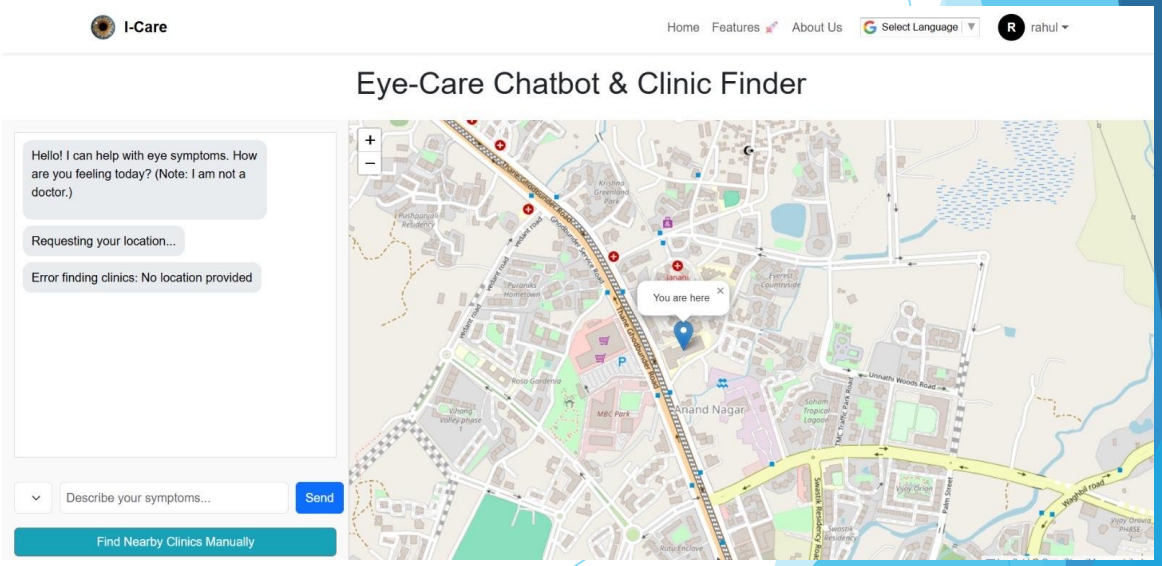

Figure 8: chatbot & clinic finder

Review Suggestions (Given in Last meeting)

1. Suggestion:
Having only 10 questions may reduce accuracy and allow random guessing.

Action Taken:
The test uses 10 carefully selected Ishihara plates to ensure quick screening with sufficient accuracy.
Results are evaluated based on overall response patterns, making it suitable for preliminary assessment, not diagnosis.

2. Suggestion:
If retinal images are taken in hospitals, diagnosis can already be done by doctors.

Action Taken:
The feature is used as a decision-support and early screening tool, providing instant preliminary results to assist doctors and support telemedicine and rural eye camps.
Final diagnosis is always left to medical professionals.

Research paper Status

The research paper is around **80%** complete. Further improvements will be made based on the suggestions provided in the review. The remaining changes will focus on refining the methodology, results, and overall presentation before final submission.

Patent/Copyright filing Status

The copyright process has not been initiated yet. We will initiate the process shortly.

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Thank You...!!